



JERUSALEM COLLEGE OF ENGINEERING

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING QUESTION BANK

Regulation	:	2023
Course Name	:	INTRODUCTION TO BIOMETRIC SYSTEMS
Course Code	:	JBM9204
Year / Semester / Branch	:	III / V / CSE (AI&ML)
Unit No	:	I
Unit Title	:	BIOMETRIC SYSTEMS AND AUTHENTICATION TECHNOLOGIES
No. of Hours	:	09
Weightage in Exam (%)	:	20

PART A

Q.No.	Questions	Marks	BTL	CO
1	Define the term biometrics with specific examples. Biometrics: Automated method of identifying or verifying the identity of a living person based on unique biological or behavioral characteristics. A biometric is a physiological or behavioral characteristic of a human being that can distinguish one person from another and that theoretically can be used for identification or verification of identity. Examples: Psychological: Iris, Fingerprints, Hand, Retinal and Face recognition Behavioral: Voice, Typing pattern, Signature Biometrics: The word "biometrics" is derived from the Greek words 'bios' and 'metric' which means life and measurement respectively, translated into "life measurement". Biometrics (ISO definition): A measurable physical characteristic or personal behavior trait used to recognize the identity or verify the claimed identity of an enrollee.	2	K1	CO1
2	Define biometric system and give the main functions of a biometrics system. • A biometric system is a combined hardware/software system for biometric identification or verification. Main functions of a biometric system are as follows: • Receive biometric samples from an enroller or candidate. • Extract biometric feature from the sample. • Compare the sample of the candidate with stored templates from individuals. • Indicate identification or verification of identity has been achieved	2	K1	CO1



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3	Define biometric authentication. Biometric authentication is a type of system that relies on the unique biological characteristics of individuals to verify identity for secure access to electronic systems. Different types of biometric authentication are • Identification (one-to-many) • Verification (one-to-one) • Watch List(one-to-few)	2	K1	CO1												
4	List out the benefits of biometric technology. • Allows rapid identification of criminals, combatants or terrorists • Enhances homeland security and fights crime • Convenient way to access secure facilities or information systems • Nothing to remember • Can't be guessed, stolen, shared, lost, or forgotten • Prevents impersonation protects against identity theft • Higher degree of non-repudiation • Enhances privacy protects against unauthorized access to personal information	2	K1	CO1												
5	Compare Identification Vs. Authentication. <table> <tr> <th>Identification</th> <th>Authentication.</th> </tr> <tr> <td>It determines the identity of the person.</td> <td>It determines whether the person is indeed who he claims to be.</td> </tr> <tr> <td>No identity claim</td> <td>Identity claim from the user</td> </tr> <tr> <td>Many-to-one mapping.</td> <td>One-to-one mapping.</td> </tr> <tr> <td>Computational cost depends on the number of user records.</td> <td>Computational cost is independent of the number of user records.</td> </tr> <tr> <td>Captured biometric signatures come from a set of known biometric feature stored in the system.</td> <td>Captured biometric signatures may be unknown to the system.</td> </tr> </table>	Identification	Authentication.	It determines the identity of the person.	It determines whether the person is indeed who he claims to be.	No identity claim	Identity claim from the user	Many-to-one mapping.	One-to-one mapping.	Computational cost depends on the number of user records.	Computational cost is independent of the number of user records.	Captured biometric signatures come from a set of known biometric feature stored in the system.	Captured biometric signatures may be unknown to the system.	2	K2	CO1
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6	State the need of Biometric system. • As the level of security breaches and transaction fraud increases, the need for highly secure identification and personal verification technologies is becoming apparent. • Biometrics based solutions are able to provide for confidential financial transactions and personal data privacy. • The need for biometrics can be found in federal, state and local governments in the military and in commercial applications. • Enterprise-wide network security infrastructures,	2	K1	CO1												



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	government IDs, Secured electronic banking investing and other financial transactions, retail sales, law enforcement, health and social services are already benefiting from these technologies.			
7	Is DNA a biometric? If so, why? Yes, because - Identity information from every cell in the body in a digital form - Not practical for everyday authentication or verification only for very high stakes - Criminal forensics and paternity	2	K2	CO1
8	State few applications of biometric at government sector level. - National ID card (e.g. Voting, Aadhar) - Receiving government benefits (e.g. welfare, pension) - Health services (e.g. Patient ID) - Law enforcement (e.g. Driver's license, vehicle registration, smart guns, criminal identification system, Border control) - Identification documents (e.g. Visa, Passport, SIN cards, Military/Govt/ Corporate ID cards)	2	K1	CO1
9	Define matching. The process of comparing an input image with biometric templates in the database to determine their degree of similarity or correlation called matching. The process of matching biometric templates results in a score, which in most systems, is compared against a threshold. If the score exceeds the threshold, the result is a match, if the score falls below the threshold, the result is non-match.	2	K1	CO1
10	List the privacy and discrimination issues in biometrics. Biometrics hacks may have greater consequences- Biometric reveals the part of an individual's identity, if it gets stolen, then it can be used to falsify legal documents, passports, or criminal records, which can do more damage than a stolen credit card pin or number.	2	K1	CO1
11	Give the ideal characteristics (Quantitative Measures) of good biometric system. Permanence (Robust): this is measured by the "false non-match rate also known as "Type 1 Error". The probability that a submitted sample will not match the enrolled image. Universality: is measured by the "failure to enroll" rate the probability that a user will not be able to supply a readable measure to the system upon enrollment. Uniqueness (Distinctive): is measured by the "false match rate" also known as Type II error". The probability that a submitted sample will match the enrolled image of another user. Collectability (Measurability): can be quantified by the	2	K1	CO1



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	“throughput rate” of the system, the number of individuals that can be processed in a unit time, such as a minute or an hour. Acceptability is measured by polling the device users. The first four qualities are inversely related to their above measures. A higher “false non match rate”, for instance indication a lower level of robustness.			
12	Define accuracy. The accuracy of biometric-like identity authentication is due to the genuine and fraud distribution of matching. The overall accuracy can be illustrated by Receiver Operation Characteristics (ROC) curve, which shows the dependence of False Reject Rate (FRR) on False Accept Rate (FAR) at all thresholds. When the parameter changes, FAR and FRR may yield the same value, which is called Equal Error Rate (EER). It is a very important indicator to evaluate the accuracy of the biometric system. The system accuracy can be improved by collecting multiple templates with different variation.	2	K1	CO1
13	Give the basic measures of accuracy of a biometric system. The basic measures of the accuracy are False Non-Match Rate (FNMR) and False Match Rate (FMR). In the context of biometric verification, FNMR and FMR are also known as False Reject Rate (FRR) and False Accept Rate (FAR), respectively. In biometric identification, the false match and false non-match errors are measured in terms of the False Positive Identification Rate (FPIR) and False Negative Identification Rate (FNIR) respectively.	2	K1	CO1
14	List out the factors required for accuracy. • Human factor: Capture of biometric information always needs human participation on different degrees. • Fingerprint: Finger must be in contact with sensor surface • Iris image: Eye must be in contact with sensor surface • Device factor: Environment and technological factors affect quality • Finger print: Light condition vary internally and externally • Algorithm factor	2	K1	CO1
15	State the benefits of biometrics versus traditional authentication systems. • Improved security – Biometric traits are difficult to reproduce or guess compared with passwords. • Convenience – Users do not need to remember passwords or carry authentication cards. • Non-transferable identity – Biometrics are inherently linked to the individual.	2	K1	CO1



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	• Reduced password-related problems – Eliminates password forgetting, reuse, and frequent password-reset requirements. • Fast authentication – Fingerprint, face, or iris recognition can provide rapid identity verification. • Resistance to credential theft – Stealing a password alone is not sufficient when biometric verification is required. • Better user accountability – Authentication is directly associated with the person rather than a shared credential. • Useful for high-security applications – Biometrics are widely applicable in banking, healthcare, border control, mobile devices, and access-control systems.			
16	State the difference between Physiological and behavioural characteristic of biometrics. • Physiological or Biological characteristics are related to the shape of the body (physical attributes of a person) such as fingerprint, palm veins, face recognition, DNA, palm print, hand geometry, iris recognition, retina and odor/scent. This is based on the fact that these person's attribute do not change over time. • Behavioral characteristics are related to behavior pattern of a person such as typing rhythm, gait and voice. This behavior is recorded in a period of time. It is very difficult to detect and emulate because it depends on an interaction of users with their own devices to extract specific and accurate habits.	2	K1	CO1
17	Define physical and logical access in the context of biometric system. • Physical access is a term in computer security that refers to the ability of people to physically gain access to a computer system • Logical access controls are tools and protocols used for identification, authentication, authorization, and accountability in computer information systems. Logical access is often needed for remote access of hardware. Militaries and governments use logical access biometrics to protect their large and powerful networks and systems which require very high levels of security.	2	K1	CO1
18	List the important factors in sensing. • Application dependent factors in sensing: User acceptance, portability, electrical and optical characteristics, resolution and price. • The surrounding environment such as vibration, temperature, electromagnetic radiation, surrounding light is affecting the sensing of the biometric system. • The sensing mechanism influences signal to noise ratio of a	2	K1	CO1



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	biometric system.			
19	Differentiate active and passive biometrics. Active Biometrics: It requires active participation of subject. Biometrics such as iris, face (controlled acquisition), speech, finger/hand prints. The system is expected to use the biometrics to confirm if the person is truly whom he/she claims to be. Passive biometrics: In passive biometrics systems the subject does not have to take active part in the process of identification verification or, in fact, does not even know that the process of identification takes place. In such biometrics the user does not have to cooperate with the system and does not need to touch any device or perform any action.	2	K2	CO1
20	Define biometric data privacy. - Biometric data is personal. - Biometric systems record personal information about identifiable individuals. That means their use by the federal government falls under the provisions of the Privacy Act.	2	K1	CO1

PART B

Q.No.	Questions	Marks	BTL	CO
1	Explain and analyze the complete architecture of a biometric system. Discuss how each functional block affects the accuracy, security, and reliability of the overall system.	16	K4	CO1
2	Explain and analyze authentication, verification, and identification processes in biometric systems. Compare their objectives, matching mechanisms, and applications with suitable examples.	16	K4	CO1
3	Explain and analyze the major physiological and behavioral biometric characteristics based on universality, uniqueness, permanence, collectability, performance, acceptability, and circumvention.	16	K4	CO1
4	Explain and analyze the privacy and security issues associated with biometric systems. Discuss biometric template protection, spoofing attacks, unauthorized access, data sharing, and privacy policies.	16	K4	CO1
5	Explain and analyze FAR, FRR, EER, and ROC curves as performance measures of biometric systems. Discuss how changing the decision threshold affects system performance.	16	K4	CO1
6	Explain and analyze traditional authentication, biometric authentication, and multi-factor authentication with respect to security, convenience, privacy, cost, and reliability.	16	K4	CO1



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Course Name	:	INTRODUCTION TO BIOMETRIC SYSTEMS
Course Code	:	JBM9204
Year / Semester / Branch	:	III / V / CSE (AI & ML)
Unit No.	:	II
Unit Title	:	FINGERPRINT IDENTIFICATION TECHNOLOGY
No. of Hours	:	09
Weightage in Exam (%)	:	20

PART A

Q.No.	Questions	Marks	BTL	CO
1	How are fingerprints captured? Two main technologies used to capture image of the fingerprint Optical- use light refracted through a prism CMOS capacitance: Capacitive based detect voltage changes in skin between ridges and valleys. Other methods: Thermal sensing and Ultrasound sensing	2	K1	CO2
2	What is meant by interoperability? It means the ability of health team work together within and across organizational boundaries in order to advance the effective delivery of healthcare for individuals and communities.	2	K1	CO2
3	How banking security is performed using finger scan? Fingerprint and Touch ID Sign-In: Banks are fully capitalizing on Apple's Touch ID and its Android counterparts found everywhere and in mobile phone feature as an added layer of security. Customers can use the fingerprint scanner on their mobile phones to sign in to the mobile banking app. The new feature makes great use of an emerging trend and reduces friction for the customer	2	K1	CO2
4	Outline the components of finger scan. • Scanning • Digitizing • Enhancement • Feature extraction • Classification • Matching	2	K1	CO2
5	Summarize the strength of finger scan. • High accuracy • Ease use	2	K2	CO2



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	- Ability to enroll multiple fingers - Minimum environment impacts			
6	Describe the merits of fingerprint identification. - Universality–Fingerprint is universally available with every individual. Only some people do not have fingers - Uniqueness–Each individual has a unique fingerprint. No two people have same fingerprint patterns - Permanence–Fingerprint remains permanently with the user right from the development of seven months fetus until the person dies - Biometrics cannot be forgotten, lost, duplicated or stolen - It is more secure as it cannot be shared or used by others - No need to remember passwords or any PINs - Physical human characteristics are very much difficult to forge than passwords, security codes, or even some encryption keys - Biometrics gives the promise of fast, easy-to-use, accurate, reliable, and less expensive authentication for many applications	2	K2	CO2
7	What are the 7 main types of fingerprint patterns? Fingerprint patterns. From top left to bottom right: loop, double loop, central pocket loop, plain whorl, plain arch, and tented arch.	2	K2	CO2
8	What is finger scan? Finger scanning also called finger print scanning. It is Ink-less Method. Process of electronically sense the ridges on a human fingers and storing as images. Ink & paper is the oldest way.	2	K1	CO2
9	What are the commercial applications based on fingerprints? Fingerprints are used in Online transaction for banking applications, used to prevent unauthorized access to ATMs, cellular phones, smart cards, desktop PCs, workstations and computer networks.	2	K1	CO2
10	Outline about fingerprints. - The fingerprint is the oldest known biometric identifier that has been used for authentication and identification purposes. Human fingerprints are detailed, nearly unique, difficult to alter, and durable over the life of an individual, making them very suitable as long-term markers of personal identity. - It is therefore natural that a wide range of fingerprint recognition systems exist for use in high security applications and for the automated identification of individuals. The convenience, security and performance characteristics have made fingerprint recognition the most widely used biometric authentication technology today.	2	K1	CO2
11	What is fingerprint technology? Fingerprint technology is the most widely deployed biometric	2	K2	CO2



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	technology, with a number of different vendors offering a wide range of solutions. Its maturity, providing a high level of recognition accuracy The growing market of low-cost small-size acquisition devices, allowing its use in a broad range of applications, e.g., electronic commerce, physical access, PC logon, etc. The use of easy-to-use, ergonomic devices, not requiring complex user-system interaction			
12	How AFIS differ from finger scan? Both Automated Fingerprint Identification Systems (AFIS) and Biometric Fingerprint Systems (BFS) are now used more as identifying individuals. AFIS was developed by the police department for identifying persons from over a large record of files. On the other hand, BFS was developed for business purposes that replaced ID cards, passwords and other methods used for identifying. AFIS mainly deals with large-scale identification that is signaling out a person from millions of others. BFS is a one to one identifying and it does not involve searching millions of the records	2	K1	CO2
13	What was the early effort taken by William Herschel and Francis Galton? Sir William James Herschel is credited with being the first European to note the value of fingerprints for identification. He recognized that fingerprints were unique and permanent. Herschel documented his own fingerprints over his lifetime to prove permanence. He was also credited with being the first person to use fingerprints in a practical manner. As early as the 1850s, working as a British officer for the Indian Civil Service in the Bengal region of India, he started putting fingerprints on contracts.	2	K1	CO2
14	What are the drawbacks of finger print authentication? A number of weaknesses may influence the effectiveness of fingerprint recognition in certain cases: Its association with forensic or criminal applications. Factors such as finger injuries or manual working can result in certain users being unable to use a fingerprint-based recognition system, either temporarily or permanently. Small-area sensors embedded in portable devices may result in less information available from a fingerprint and/or little overlap between different acquisitions.	2	K2	CO2
15	What is the term feature extraction means in finger prints scans? Features extracted from a fingerprint image are generally categorized into three levels. Level 1: features capture macro details such as friction ridge flow, pattern type, and singular points. These features can be used to categorize fingerprints into major pattern types such as arch, loop, or whorl. Level 2 features refer to minutiae such as ridge bifurcations and endings. Level 3 features include all dimensional attributes of the	2	K2	CO2



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	ridge such as ridge path deviation, width, shape, pores, edge contour, and other details, including incipient ridges, creases, and scars.			
16	Explain welfare fraud reduction. Welfare fraud is the act of illegally using state welfare systems by knowingly withholding or giving information to obtain more funds than would otherwise be allocated. Obtaining reliable evidence of welfare fraud is notoriously difficult. Official figures of the prevalence of welfare fraud based on government investigation tend to be low – a few percent of the total amount of welfare spending. Welfare fraud involves modest sums and is committed by people who struggle with poverty, but once started it often continues after reaching financial stability.	2	K2	CO2
17	What is Welfare Fraud? Recipients of public assistance are committing fraud if they: - falsely report that a parent is absent from the household - intentionally fail to report a change in the household composition, income or resources that would affect the eligibility for benefits - intentionally fail to report any change in circumstances that would affect their eligibility for benefits - Intentionally receive benefits from two or more authorities at the same time - intentionally receive benefits for children not in the household.	2	K1	CO2
18	What are the main objectives of contextual filtering used to enhance fingerprint patterns? - A single fingerprint image contains regions of good, medium, and poor quality, where the ridge pattern is very noisy and corrupted. - Techniques such as gray-level smoothing, contrast stretching, histogram equalization can be used as pre-processing steps before a sophisticated fingerprint enhancement algorithm is applied. - The estimation of orientation field plays an important role in the fingerprint enhancement. - Good fingerprint enhancement techniques obtain a relatively good estimate of orientation field even if the quality of input fingerprint image is poor. - Most of the fingerprint enhancement techniques use contextual filter or multi-resolution filter. They used filtering masks similar to the ones used to estimate orientation, either fixed size or predetermined variable frequencies	2	K2	CO2
19	How fingerprint identification is performed? The process contains four main steps as follows: 1. Fingerprint Image Acquisition 2. Image Processing 3. Feature extraction	2	K2	CO2



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	4. Template Creation 5. Template Matching			
20	Do biometric features remains constant with time? The iris is a visible but protected structure, and it does not usually change over time, making it ideal for biometric identification. Most of the time, people's eyes also remain unchanged after eye surgery, and blind people can use iris scanners as long as their eyes have irises. DNA biometric features are also remains constant with time.	2	K2	CO2

PART B

Q.No.	Questions	Marks	BTL	CO
1	Explain and analyze the major stages of a fingerprint recognition system and discuss the role of each stage in improving recognition accuracy.	16	K4	CO2
2	Explain and analyze the working principle of RF-based fingerprint sensors	16	K4	CO2
3	Explain and analyze the problems affecting fingerprint image quality and discuss suitable enhancement techniques to overcome them.	16	K4	CO2
4	Explain and analyze the minutiae extraction process in fingerprint recognition	16	K4	CO2
5	Explain the different techniques used for computer enhancement and modelling of fingerprint images.	16	K3	CO2
6	Explain and analyze fingerprint enhancement and modelling techniques.	16	K4	CO2



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Year / Semester / Branch	:	III / V / CSE (AI & ML)
Unit No	:	III
Unit Title	:	FACE RECOGNITION AND HAND GESTURE RECOGNITION
No. of Hours	:	09
Weightage in Exam (%)	:	20

PART A

Q.No.	Questions	Marks	BTL	CO
1	Define face recognition. Facial recognition is a way of identifying or confirming an individual's identity using their face. Facial recognition systems can be used to identify people in photos, videos, or in real-time.	2	K1	CO3
2	List the various applications of face recognition technology. Security & Surveillance – Monitoring public places, airports, and high-security zones for identifying persons of interest. Access Control – Replacing passwords or keycards for secure entry into buildings or systems. Law Enforcement – Locating missing persons or suspects using CCTV footage and criminal databases. Border Control & Immigration – Verifying travelers' identities at airports and checkpoints. Smartphones & Personal Devices – Unlocking devices, authorizing app logins, or making secure payments. Attendance Systems – Automatic marking of attendance in workplaces, schools, or events. Banking & Financial Services – Identity verification for secure transactions and fraud prevention.	2	K1	CO3
3	Define Eigenface. - An Eigenface is a set of standardized face components (features) that are derived from a large set of facial images using Principal Component Analysis (PCA). - PCA finds the patterns of variation (like common shapes, shadows, and structures) across all those faces. These patterns are represented as Eigenfaces — basically, "face templates" that capture the main differences between faces. - Each face can then be represented as a combination of	2	K1	CO3



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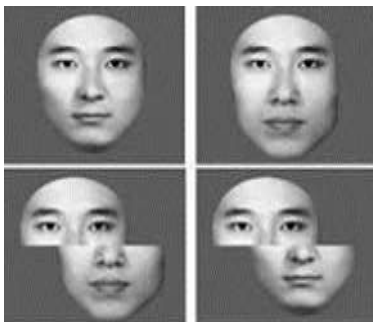
	Eigenfaces, instead of storing the full image.			
4	Explain FERET database. The most frequently used dataset is developed for FERET program. The FERET database is a dataset used for facial recognition system evaluation. The Face Recognition Technology (FERET) program is managed by the Defense Advanced Research Projects Agency (DARPA) and the National Institute of Standards and Technology (NIST). A database of facial imagery was collected between December 1993 and August 1996. In 2003 DARPA released a high-resolution, 24-bit color version of these images. The dataset tested includes 2,413 still facial images, representing 856 individuals.	2	K2	CO3
5	Differentiate between face detection and face recognition. • Face detection identifies the presence and location of a face in an image. • Face recognition matches the detected face to a stored identity.	2	K2	CO3
6	Identify the types of facial recognition. Facial recognition algorithms can be broadly classified into two types: 1. Holistic (Statistical) models – These analyze the entire face as a whole for recognition. 2. Feature-based (Geometric) models – These focus on individual facial components (such as eyes, nose, and mouth) and examine both their characteristics and their spatial relationships to each other.	2	K1	CO3
7	State the pioneers of face recognition. The earliest pioneers of facial recognition were Woody Bledsoe, Helen Chan Wolf and Charles Bisson. In 1964 and 1965, Bledsoe, along with Wolf and Bisson began work using computers to recognize the human face.	2	K1	CO3
8	Is face recognition a skill? Although researchers have historically examined people with a very high IQ (i.e. those with more generalized skills), it has become more and more clear that some individuals only perform extraordinarily well on specific cognitive tasks. Among the most interesting of these is facial identity recognition. 	2	K1	CO3
9	Explain genetic influence on face recognition ability.	2	K2	CO3



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	Researchers found that identical twins — who share all of their genes — are more than twice as alike in their face recognition ability compared to fraternal twins, who share roughly half of their genes. This indicates a strong genetic influence on face recognition skills.				
10	Explain the role of training in neural networks for face recognition. It adjusts the network’s weights using face image data so the model can learn to accurately extract features and classify or recognize faces.	2	K2	CO3	
11	List out the feature extraction in hand geometry. Feature extraction in hand geometry is the process of measuring and deriving unique, quantifiable characteristics from a person’s hand that can be used for biometric identification or verification. After the hand image is captured and segmented, feature extraction focuses on identifying measurable traits such as: • Finger lengths (distance from fingertip to base) • Finger widths (measured at specific points) • Palm width and length • Shape of the hand • Distances between joints • Aspect ratio of fingers	2	K1	CO3	
12	Define hand geometry. Hand geometry recognition is a biometric approach to automatically recognize individuals based on the unique geometric features of the hand.	2	K1	CO3	
13	Difference between palm scan and hand geometry. Palm Scan: The palm has creases, ridges and grooves throughout it that are unique to a specific person. Hand Geometry: The shape of a person’s hand (the length and width of the hand and fingers) measures hand geometry.	2	K2	CO3	
14	List the advantages of hand geometry. • Universality. • Uniqueness. • Permanence. • Collectability Performance. • Acceptability.	2	K1	CO3	
15	Define hand segmentation. Hand segmentation is the process of isolating the hand region from the rest of the image so that only the relevant part is analyzed for	2	K1	CO3	



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Approved by AICTE, New Delhi, Affiliated to Anna University Chennai,
Accredited by NBA, New Delhi and Accredited by NAAC with "A" Grade.
Velachery Main Road, Narayanapuram, Pallikaranai, Chennai, Tamil Nadu - 600100.

	biometric recognition or gesture processing. It is an essential preprocessing step in hand geometry recognition and hand-based biometric systems because accurate segmentation ensures precise feature extraction.			
16	Define adaptive classifier. An adaptive classifier is a model that updates its parameters or decision boundaries dynamically using new incoming data (online or incremental learning) so it can track changes in the data distribution (concept drift).	2	K1	CO3
17	Define visual-based feature extraction. Visual-based feature extraction is the process of obtaining discriminative information directly from the appearance of an image (such as texture, shape, or intensity patterns) without relying on geometric measurements. Techniques like Principal Component Analysis (PCA), texture analysis, and Gabor filters are commonly used to capture these visual patterns for tasks like face or hand geometry recognition.	2	K1	CO3
18	Define pattern classification? Pattern classification is the process of assigning input data (features extracted from images or signals) into predefined categories or classes based on their characteristics. It involves using algorithms like k-Nearest Neighbor (k-NN), Support Vector Machines (SVM), or Neural Networks to recognize patterns and make decisions for applications such as biometric identification.	2	K1	CO3
19	List the challenges faced in hand-based biometric recognition systems. Hand-based biometric recognition systems face challenges such as variations in hand positioning and orientation during image capture, changes in lighting conditions affecting image quality, and difficulty in segmenting the hand accurately from complex backgrounds. Additionally, factors like skin texture changes, injuries, or wearing accessories (rings, bracelets) can reduce recognition accuracy.	2	K1	CO3
20	List out some common algorithms used for hand segmentation in biometric systems. Common algorithms for hand segmentation include • Skin color-based segmentation (using color spaces like HSV or YCbCr) • Thresholding techniques based on pixel intensity • Edge detection methods such as Canny or Sobel operators, and • Background subtraction to isolate the hand from static backgrounds.	2	K1	CO3



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PART B

Q.No.	Questions	Marks	BTL	CO
1	Explain and analyze the complete face recognition system with its major processing stages.	16	K4	CO3
2	Explain and analyze different face recognition algorithms and their suitability.	16	K4	CO3
3	Explain and analyze neural networks for face recognition.	16	K4	CO3
4	Explain the complete hand geometry recognition system with image capturing, segmentation, feature extraction and classification.	16	K4	CO3
5	Explain and analyze hand segmentation and feature extraction techniques.	16	K4	CO3
6	Explain and analyze adaptive classifiers for hand geometry recognition.	16	K4	CO3
7	Explain and analyze visual-based feature extraction and pattern classification.	16	K4	CO3
8	Explain and analyze face recognition and hand geometry recognition comparatively.	16	K4	CO3